

Engineering Electromagnetics II
Homework #4 Solutions
48 points total

ECE 332 – Fall 2025

Due Friday, December 5, 2025 at 10 pm

1. A 1-m long dipole is excited by a 1 MHz current with an amplitude of 12 A. What is the average power density radiated by the dipole at a distance of 5 km in a direction that is 45° from the dipole axis? (8 points)

At 1 MHz,

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{10^6 \text{ s}^{-1}} = 300 \text{ m}.$$

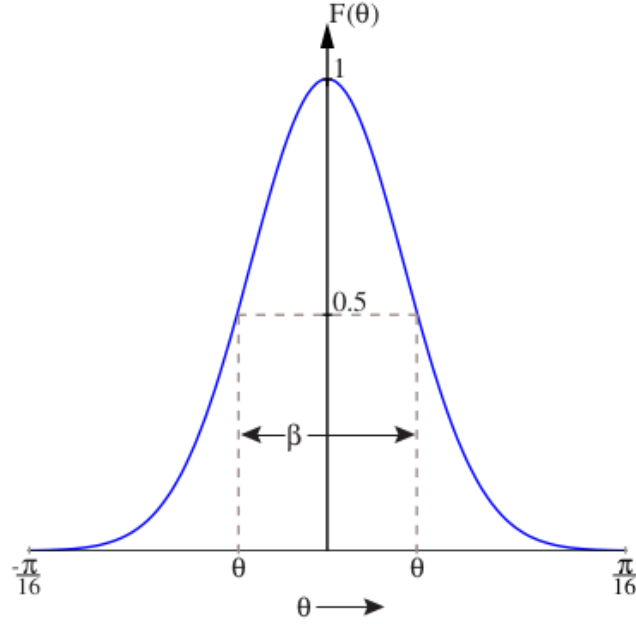
So, $l/\lambda = (1 \text{ m})/(300 \text{ m}) < 1/50$, which means that the antenna is a Hertzian dipole. Therefore,

$$\begin{aligned} S(R, \theta) &= \left(\frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \right) \sin^2 \theta \\ &= \frac{(120\pi)(2\pi/300)^2(12)^2(1)^2}{32\pi^2(5 \times 10^3)^2} \sin^2\left(\frac{\pi}{4}\right) = 1.51 \times 10^{-9} \text{ W/m}^2. \end{aligned}$$

2. The normalized radiation intensity of a certain antenna is given by

$$F(\theta) = \exp(-20\theta^2) \text{ for } 0 \leq \theta \leq \pi,$$

where θ is in radians. Determine: (15 points total)



- the half power beamwidth, (5 points)

Since $F(\theta)$ is independent of ϕ , the beam is symmetrical about $z = 0$. The beam has half power when,

$$\exp(-20\theta^2) = F(\theta) = 0.5.$$

This means

$$-20\theta^2 = \ln(0.5) \implies \theta = \sqrt{-\ln(0.5)/20} = \pm 0.186 \text{ rad.}$$

So,

$$\beta = 2\theta = 0.372 \text{ rad} = 21.31^\circ.$$

- the pattern solid angle, (5 points)

The pattern solid angle is

$$\begin{aligned} \Omega_p &= \int \int_{4\pi} F(\theta) \sin \theta \, d\theta \, d\phi = \int_0^{2\pi} \int_0^\pi \exp(-20\theta^2) \sin \theta \, d\theta \, d\phi \\ &= 2\pi \int_0^\pi \exp(-20\theta^2) \sin \theta \, d\theta. \end{aligned}$$

This integral can be evaluated using software to determine

$$\Omega_p = 0.156 \text{ sr.}$$

- the antenna directivity. (5 points)

Here,

$$D = \frac{4\pi}{\Omega_p} = \frac{4\pi}{0.156} = 80.55.$$

3. For a dipole antenna of length $l = \lambda/4$, (15 points total)

- determine the directions of maximum radiation, (5 points)

For an arbitrary length dipole,

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left(\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right)^2.$$

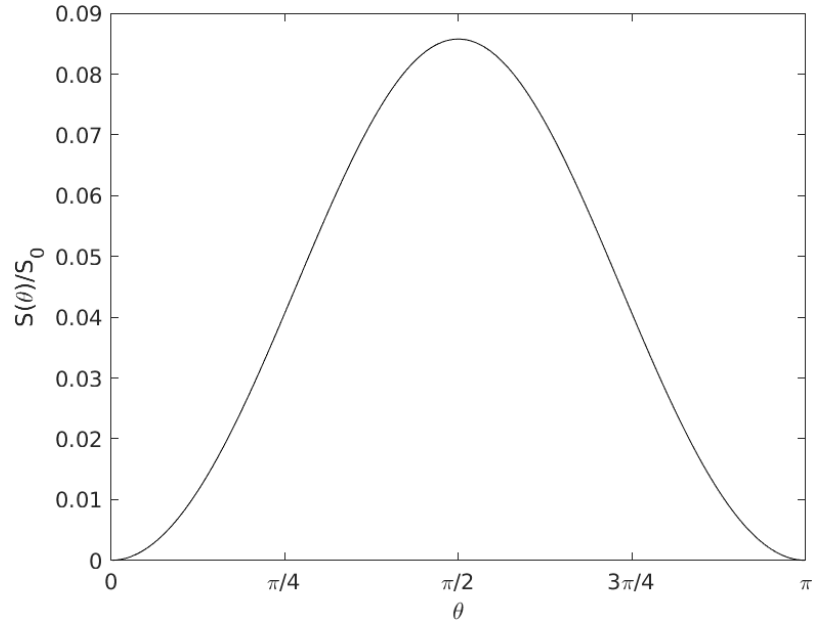
When $l = \lambda/4$, $\pi l/\lambda = \pi/4$ and

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left(\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \cos\left(\frac{\pi}{4}\right)}{\sin \theta} \right)^2 = \frac{15I_0^2}{\pi R^2} \left(\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right)^2.$$

For given values of I_0 and R , the factor $S_0 = 15I_0^2/(\pi R^2)$ is a constant and maximizing $S(\theta)$ with respect to θ means maximizing

$$\frac{S(\theta)}{S_0} = \left(\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right)^2$$

for $0 \leq \theta \leq \pi$. A plot of $S(\theta)/S_0$ is shown below.



The usual methods from calculus or numerical optimization show that $S(\theta)$ reaches its maximum value at $\theta = \pi/2$.

- obtain an expression for $S_{\max}$, (5 points)

The average power density reaches its maximum value $S_{\max}$ at $\theta = \pi/2$, i.e.

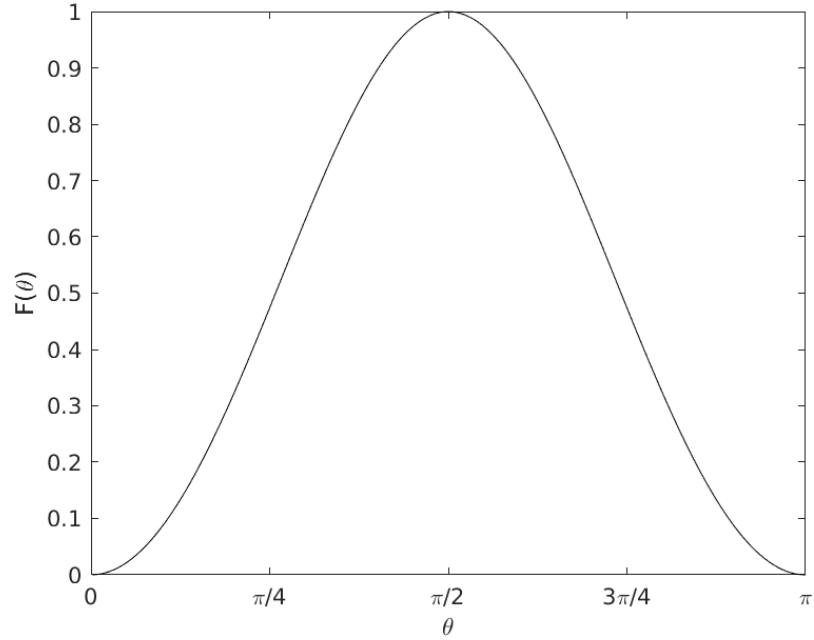
$$\begin{aligned}
 S_{\max} = S(\pi/2) &= \frac{15I_0^2}{\pi R^2} \left(\frac{\cos\left(\frac{\pi}{4} \cos\left(\frac{\pi}{2}\right)\right) - \frac{\sqrt{2}}{2}}{\sin\left(\frac{\pi}{2}\right)} \right)^2 \\
 &= \frac{15I_0^2}{\pi R^2} \left(\frac{\cos(0) - \frac{\sqrt{2}}{2}}{1} \right)^2 = \frac{15I_0^2}{\pi R^2} \left(1 - \frac{\sqrt{2}}{2} \right)^2 \\
 &= 0.0858 \left(\frac{15I_0^2}{\pi R^2} \right).
 \end{aligned}$$

- generate a plot of the normalized radiation pattern $F(\theta)$. (5 points)

In this case, the normalized radiation pattern is

$$F(\theta) = \frac{S(\theta)}{S_{\max}} = \frac{1}{0.0858} \left(\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right)^2,$$

which is shown below.



4. Determine the effective area of a half-wave dipole antenna at 100 MHz, and compare it with its physical cross section if the wire diameter is 2 cm. (5 points)

In this case, $f = 100$ MHz and

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{10^8 \text{ s}^{-1}} = 3 \text{ m}.$$

The directivity of a half-wave dipole antenna is $D = 1.64$. So, the effective area is

$$A_e = \frac{\lambda^2 D}{4\pi} = \frac{1.64(9 \text{ m}^2)}{4\pi} = 1.17 \text{ m}^2.$$

By comparison, the physical cross section is

$$A_p = ld = \frac{1}{2}\lambda d = \frac{1}{2}(3 \text{ m})(2 \times 10^{-2} \text{ m}) = 3 \times 10^{-2} \text{ m}^2.$$

So,

$$\frac{A_e}{A_p} = \frac{1.17}{3 \times 10^{-2}} = 39.2.$$

5. A half-wave dipole TV broadcast antenna transmits 1 kW at 50 MHz. What is the power received by a home television antenna with a 3 dB gain if located at a distance of 30 km? (5 points)

Here, $f = 5 \times 10^7 \text{ s}^{-1}$ and

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{5 \times 10^7 \text{ s}^{-1}} = 6 \text{ m}.$$

If the transmitter is a vertical half-wave dipole antenna, then its gain is $G_t = 1.64$. The gain of the receiving antenna is

$$G_r = 3 \text{ dB} = 10^{3/10} = 2.$$

Assuming that the home antenna and the transmitting antenna are properly aligned,

$$\begin{aligned} P_{\text{rec}} &= G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t = (1.64)(2) \left(\frac{6 \text{ m}}{4\pi(3 \times 10^4 \text{ m})} \right)^2 (10^3 \text{ W}) \\ &= 8.3 \times 10^{-7} \text{ W}. \end{aligned}$$